

Cosmic Queries – Pure Spacetime

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis The episode explores the fundamental, non-absolute nature of the universe through a series of cosmological and physical inquiries. The central theme is that while phenomena such as time dilation, gravity, and the "cosmic web" may appear chaotic or even "programmed," they are governed by consistent, measurable laws of physics—such as relativity, quantum mechanics, and electromagnetism—that allow for scientific predictability and the pursuit of objective truth.

Key Arguments & Data

- * **General Relativity and Time Dilation:** Time is not an absolute constant; it varies based on velocity and gravitational intensity. Near a black hole, an observer would see time in a different gravitational well proceeding at a different pace (e.g., seeing someone move in "slow motion"). This principle is practically applied today via GPS satellite corrections, which must account for the difference in time between Earth's surface and orbit.
- * **The Simulation Hypothesis:** The debate over whether we live in a simulation hinges on the existence of consistent "laws." If the universe follows predictable patterns (physics), science remains a valid tool for discovery regardless of whether those laws are natural or programmed by an "overlord."
- * **Quantum Superconductivity:** At extremely low temperatures (near 3 Kelvin), electron wavelengths can overlap and synchronize, allowing electricity to flow without resistance. While possible in deep space, proximity to stars typically raises asteroid temperatures above this threshold.
- * **Lunar and Solar Tidal Forces:** The Moon is the primary driver of Earth's tides, but the Sun contributes approximately one-third of the tidal force. High tides are maximized during Full and New Moons when the Sun, Earth, and Moon align.
- * **Neurobiology and Consciousness:** Human consciousness, memory, and identity are products of electrochemical signals within the brain's neural pathways. Physical degradation of brain tissue (via strokes or dementia) directly results in the loss of specific cognitive functions and personality traits, demonstrating that "mind" is tied to biological structure.
- * **Black Hole Physics and Information Theory:** A major debate exists regarding whether information falling into a black hole is destroyed. The principle of Hawking radiation suggests that information must be preserved (though

perhaps altered) to maintain the integrity of physics. Furthermore, black holes are not "suction machines" but objects of extreme surface gravity; if the Sun became a black hole, Earth's orbit would remain relatively unchanged. * **The Cosmic Web:** Galaxies are not distributed randomly or evenly across the universe. Instead, they form a "cosmic web"—a non-uniform, sponge-like structure of filaments and clusters composed largely of dark matter, separated by vast cosmic voids.

Actionable Takeaways * **Seek Patterns in Chaos:** When encountering complex or seemingly random systems (such as weather or large-scale cosmic structures), look for the underlying variables and "handles" (models) that govern the chaos rather than assuming total randomness. * **Apply Physics to Biology:** Recognize the utility of fundamental physics in medical advancement; understanding the electrochemical and atomic nature of cells is essential for diagnosing and treating neurological and physiological conditions. * **Prioritize Objective Truth over Comfort:** In both scientific inquiry and personal life, value the acquisition of all objective truths (even if they are unsettling) as the only way to fully understand and navigate the forces that control one's environment.

Full Transcript

Paul, that was a good set of questions. Oh my gosh. It's amazing. We learned a lot about sponges and webs and all sorts of things. Coming up, another installment of Cosmic Query's Grab Bag Edition. Welcome to StarTalk, your place in the universe where science and pop culture collide. StarTalk begins right now. This is StarTalk Cosmic Queries Edition. This time I got Paul McCurio. Paul, welcome back, dude. All right, good to see you again. Neil, how are you, buddy? All right. Paul, at some point, somebody christened you barren, Paul McCurio. That is true. And we had a ceremony, which was quite lovely in your office. If you remember, you... Did we use my Excalibur? Excalibur store? Yes, we did the whole thing. Yeah, I still... Very good. For some reason, there's no...

red carpet here, but you know, you still have your traveling show permission to speak. I do. Yeah. There's a theater show, right? It's it's a it's a wagon and I sell medicines. I'm a medicine man now because I'm going to be out of a job at the late show. So yes, touring with that, I've got my podcast inside out with Paul Mercurio because it's the law and everyone's supposed to have a podcast now. And you've been in the biz a while. Yeah, I go back to the Daily Show and one of the original guys on that. Even predating John Stewart. John Stewart, that tiny little man. Yes, he's only this big. A lot of people don't know. The cameras are amazing that we would have. Yes, with Craig Kilbourne and then Lewis Black was on and we would have all this crazy fun and nobody knew what they were doing but they were letting us, we were like, You know, you would appreciate this. We were like in the lab just said, just go make it and blow things up. And then we get yelled at by the president of the network every once in a while because we went too far. He would literally yell at you like your father. We yell at you like that. He couldn't get. Why did you think that that could be OK on TV? And he's like screaming at you. We got a grab bag this session. Yeah. So that means great questions, by the way, some really good questions. Whatever subjects just sort of landed in your lap.

And these are questions asked by our Patreon members. And whenever I do these, I'm always, and this isn't false praise, like you're the fans of our show,

just how smart they are about this stuff. Like they did really great questions. Oh, yeah. Yeah. Yeah. Yeah. It's like, oh my gosh. which we handed out degrees. Except for this first guy. This first guy is an idiot. No, I'm just kidding. Okay, so what you got for me? This is Farouk from Uzbekistan, writing from Norwich, UK. It's my favorite movie interstellar. After I've seen your interview with Kip Thorne, there's something that bothers me now. When they were on Miller's planet, with giant water waves next to the black hole, being immensely time-dilated relative to one of their crew that remained behind on the ship and aged faster, what if they had telescopes and looked through the windows of the ship at their crewmate, aging years for their minutes? Wouldn't they be able to see the faster passage of time or Would it have well or what would it have looked like? Or is light itself affected and won't allow this to happen So if you had a telescope and look down at them near the surface of the black hole everything would be going in slow motion I understand but it's also kind of insensitive listen with it a look into the telescope. Hey guys. Look at this Look at he's aging so fast. He just he just took out a reverse mortgage on his house He's turning gray while we watch it. Yeah, I I just had a sandwich and during that time he's been divorced three times and he doesn't have any visitation rights. Like what? They would look down and see them moving slowly and when you're near a black hole you see the rest of the universe unfold in rapid speed. And so yeah, there's nothing special about the light that happens. It arrives to you at a different pace.

depending on how deep into a gravitational well you are. And the reason why that orbit was chosen to be very close to the black hole, it was to maximize the time dilation. It's called time dilation, when you create this difference in the rate of time. By the way, that exists for our GPS satellites. We are closer to the center of the Earth than the GPS satellites are, so we experience slower time than they do, yet we get our precise time from them. So, when they send down their time from orbit, they pre-correct it for Einstein's general theory of relativity so that our time works at our location on Earth's surface. Okay. So, in saying you're looking through the telescope, you're seeing things in slow motion, essentially? Is that right? Well, if you're looking at their behavior on the planet, yeah. Oh, yeah. Because even if the guy...

because light itself gets stretched. And so it looks like he's buffering, basically. Is that what we're saying? Buffering. That's an interesting word to bring into relativity. I have to think about that. So no, just the timeline. So here's the thing. You are a prisoner of the present forever transitioning between your inaccessible past and your unknowable future. That's all you know about your own timeline. There are other timelines that are proceeding at different paces than your timeline, depending on how fast they're moving in space, relative to you, and depending on how much gravity they are subjected to. So time is not absolute. This was the big discovery of Einstein. Isaac Newton is turning in his grave upon hearing this.

If you could hear let's say who's present are we it's basically like who's present are we talking about right does this? Relativity does this mean there's no shared now So the idea of watching someone age across frames is already kind of the wrong question in a way Some of those questions you're right some of those questions just kind of lose meaning like what is the now? For everybody if your timeline is unfolding more slowly than mine Right because you can't watch them age and fast forward because the universe won't let you skip ahead to the consequence. It won't let you skip ahead, but you can watch a timeline unfold faster than yours. And if time is that dilated, does that mean you couldn't actually watch him aging and fast forward because the light carrying that information is stretched and delayed as well? So the universe... Yeah, I think if you're tracking time, the stretching of the So rather than think of stretching the light, think of light pulses, right? Got it. And so what's the interval between pulses? And that's a nice way to sort of tick out the time. The pulses you would see down on the planet, I guess it was called Miller's Planet. I didn't remember the name of it. Yes, Miller's. If you all agreed, let's pulse every second. And then you see their pulses down there, and their pulses happen every two or three seconds, even though according to their watches, It's every one second. So it's not that the light stretches so much as the pace at which time unfolds shifts. And so, yeah, and they did it right. We did it right. Mark Phillips, Florescent, Missouri. Hi, Dr. Tyson. This is Mark Phillips. Florescent? Florescent, F-O-F-L-O-R-I-S-S-A-N, or Florescent. If we are living in a simulation, would you want to know?

And would you choose to embrace your place or attempt to break the cycle and communicate to the overlords knowing you, Neil, you'll need to communicate because it's got a lot of that? I love that question. And third part of the question, what would you wish to gain from contact with them or have you already and you are just controlling us? I'm not authorized to comment. That's a great question. I love it. I embrace all objective truths, no matter what. I mean, it's not that long ago in the era of our grandparents, perhaps, where it was perfectly acceptable for the doctor to lie to you to tell you you don't have terminal cancer and know you're not going to be dead in six months. That was somehow accepted. That's right. In fact, I think There was such a scene in, was it a streetcar named Desire? What's the story that has Big Daddy in it? Oh, that's mendacity. That's a cat on a hot tin roof? Well, a cat on a hot tin, a cat on a hot tin, I don't know why I confused those two. So cat on a hot tin roof, Big Daddy, or whatever his name was, he, he was told, everyone else knew he was going to die and the doctor told him he was like just fine. That's right. That is to me, that is today completely unacceptable. At least for me, I want to know all truth. So that.

I can reckon with that truth in whatever way I see fit. To live a lie is to not fully be in touch with all that controls your life, all that matters. So yeah, maybe that's because I'm a scientist. Some people just say lie to me. Lie to me. They have a lot of politics, right? Lie to me. Tell me there is no global warming. Tell me when I tell me just lie to me and I know everything will be better and and that's okay Temporarily and then it catches up with you and by then it's too late Yes, so yeah, I would want to know if I was in a simulation. I I agree I would want to know be I totally want to talk to the overlords I want to talk to the manager. I have questions. You want to be Karen, right? Who's in charge? Who's in charge? First of all, who picked my avatar? Why don't I have a better body? That would be one. You had infinite rendering power and you chose to make me a guy who pulls a hamstring tying his shoe. That would be number two. Why did you give me mild acid reflux? I have a lot of issues with this overlord person.

And with our luck, it's a 14-year-old kid who gets like a triple A plus. It's an alien kid in his parents' basement. Yeah. Listen, if we're in a simulation, I just want the patch notes. That's all I want, okay? The programmer of the simulation in which we live gave you freedom to join a fitness center,

which you have in town. It is. Obvious to me, you've never stepped into a fitness area in your life. Wow. Okay. This is why I don't want to know all truths. This is why I want... There are some things I don't want to know. So what would you change? No, no, questions. What would I ask? And I would ask, is every bit of this universe knowable through laws of physics? Or is there something you just got lazy programming?

And it will never make sense to anybody and no one can ever create a law for it because you just pulled it out of your ass. That's what I'd want to know because if they didn't, then I would still continue to explore with the laws. Well, it's funny because in my mind when I read this question before we came on the air was like it's sort of maybe the other side of the coin of what you just said, which is is there only a simulation or is there some? being or power, ephemeral, ethereal, that has some hand in how we exist. I don't know that we would ever be able to deduce the difference. Well, but that's the point, right? So the simulation maybe only exists because of this greater power being, right? You'll never know the answer to that. The simulator is the greater powerful being.

Yeah, but that's deal with that. Yeah, but but we don't know what the sim is a simulator a human being Is it a is it an alien of some kind? Like what is who is the simulator? It doesn't matter. I mean I'm saying it's someone vastly Smarter than us. That's a whoa. It's like an answer Speak for yourself, buddy. Mr. Insecure low IQ guy I can catch. First of all, in this one question, you told me I have arms like wimp spaghetti noodles in terms of my body's height and I'm dumb as a stump. Is this how we're going to do the whole show? Or is it going to end here? No, I only talked about your arms. I didn't talk about your brain. You have a brilliant brain and spaghetti noodle arms. No, no, so that's what I would do. I would say, shall I continue as a scientist exploring the no ability of the universe?

Or is there some point where you just gave up and it's all random? That's what I want to know. It has good precedent. There was a day before Newton, you'd look up and the planets would move against the background stars and the moon and the sun, and no one really understood it. We could sort of trace it, but an understanding eluded us, but that was okay because the sky was the realm of God. And how could you possibly understand the

mind of God? And then Isaac Newton says, here it is, it's three equations. And some accused Newton of heresy for stepping where previously only God had tread. And so in this case, is God a deity as known through religious texts? Or is God a programmer in an alien basement?

In either case, whatever is going on is created by them and is vastly more complex than we are. And we are just inside of it, just poking... Well, what is it? I mean, if it's so sophisticated that it can create all of this, one question I would have in all seriousness is why have you created so many imperfections when you have this amazing ability to create this simulation? Okay, so I have an answer to that. I have an answer. Okay. Okay. Just when everything is going fine, just when everything is going fine, the programmer gets bored and say, let's throw in COVID. Yes. Stir the pot. I know. Let's throw in the feature where Paul has to get up three times a night to pee. How about that? Let's throw that in there just to make it like, you know. That's TMI, Paul. I don't need to know about your bladder condition.

Then, we finally get through COVID, okay? And we have the vaccine. Then, a billionaire New York real estate developer becomes the most powerful person in the world. Stare at the pot some more. If there were ever evidence that an alien programmer were in charge just for their entertainment, here's that. Hi, I'm Ernie Carducci from Columbus, Ohio. I'm here with my son, Ernie, because we listen to Star Talk every night and support Star Talk on Patreon. This is Star Talk with Neil deGrasse Tyson. So Paul, here's an interesting fact. If we reach the point where, in our exploration of the universe, where it's all random because the programmer got lazy, all right, that's interesting. At that point, science loses all utility. Science exists only because something I figure out here applies over there. And I can predict its behavior and understand it and manipulate it and use it in the service of my own survival. And if it were all random, there would be no such thing as science. But couldn't you use science to help explore and try to explain the randomness and the chaos?

Seriously. So you want a scientist to say, okay, Paul, today your whole day will be chaotic. I predict that. Go forth. What are you expecting? No, no, no, no. But can't you make the argument that you could use science to sort of explain chaos? I mean, isn't there... Not truly random. I mean, chaos is a

very specific thing. Right. And mathematically and... The interesting thing about chaos is you can predict with precision how uncertain certain quantities will become over time. And you can watch the chaotic regime unfold. This is what makes weather predicting so challenging because there's so many variables, the heat from the sun, the absorption of the earth, the reflectivity of the earth, the...

the oceans, which are a different temperature from the land, how much moisture is in the air. All of these factors come together. And the more they come together, and the more in the future you try to understand it, the more chaotic it becomes. So we try to get handles on it. That's why they say the models, we have this model and we have that model. And here's where they agree. So you go where they agree to try to boost your chances of understanding the objective reality of the system. But I'm just saying, if no laws discovered here apply over there, and it doesn't matter where here is, and it doesn't matter where there is, then that undermines the entire profession of science. And the fact that I tell you that on Earth as it is in the heavens, laws of physics on Earth apply to the moons, planets, stars, and galaxies, not only across space, but through time, that is not an assumption.

that is a measurement we have made. And that's extraordinary. Is there any reason? Is there a tablet in the sky that says that must be so? I don't know. I'm delighted we live in a universe that it is so. Absolutely. All right, let's move on to the next question. This is Mikhail from Canada. Since it's around three Kelvin in space, could an asteroid be a superconductor? And if so, would it get stuck if it... If it went too close to a huge magnetic field like the suns or ours, there you go. I love that question. Very clever. What's the person's name? This is Mikhail. Mikhail. So thank you for that question. Mikhail knows that metals, certain metals and other materials, when you reduce the temperature in which they're immersed, the behavior electron changes so that they no longer act separately.

from one another. As the temperature drops, as they get cooler, the wavelength of each electron grows bigger and bigger and bigger. Until it reaches... I'm simplifying this, but it's basically what's happening. Until the temperature is so low... By the way, this is a quantum phenomenon. Was that absolute zero? Is that absolute zero? There's three degrees above

absolute zero. So, no more heat can be taken out. When it gets to zero. So as you drop the temperature, the wavelength of the electrons gets bigger and bigger. Quantum mechanics dictates this to a point where the wavelength is so large, all the electrons behave as one electron because their waves overlap and synchronize. And when that happens, electrical current passes through without any resistance at all. And so it's called superconductivity.

At three degrees ought to get some superconductivity in some metals. I haven't checked the chart lately at who goes superconducting at what temperature. Some goes superconducting at even cooler than three degrees Kelvin. So let's start with that. So great question. I'd have to check my numbers to see who would be superconducting at those temperatures. A, B, it's not entirely three degrees if you're near a star. So the asteroids are orbiting the sun. Many of them are very dark. Sunlight gets absorbed, raising their temperature. So you're only going to hit three degrees like an intergalactic space. We are so far away from nearby stars that there's very little heat added to your rock temperature. Jason Ashton from Liverpool, Liverpool. The Beatles are from Liverpool, yeah. Wait, if he's from Liverpool, you have to read it with a Liverpool accent.

Oh, please. No, that's not deliverable. That's like Irish or something. Yeah. Oh, the home of the Beatles. Yeah. I like you. You know, I had Paul McCartney on my podcast, you know, along with you. Okay, let's move on. I've been a Patreon member for over a year and finally have a question that is probably more basic than my and your expectations are. So I'm feeling a little bit dumb here. If the Moon left Earth's orbit, would the oceans no longer have tides? They would still have tides caused by the Sun. Yes. The Sun's tides are about one third the strength of the moon's tides, or to put more precisely, whatever is the height of the tide that you measure, a third of that height is attributable to the sun. Yeah, it's smaller, less dramatic. It's like what I like to call decaf gravity. It's like decaf, yeah. That works. And what's more important here, and by the way, it's a perfectly legitimate question. I love it. And so, in fact, there's a whole book written by a colleague of mine called, What If There Were No Moon?

And then the whole book just goes, talks about the consequences of that. And we probably wouldn't have moons, because they're named after the

moon. That's true. Yeah. Yeah. We have some other sort of reckoning of time. So the moon has the dominant tidal bulge, right? So there it is. And the earth turns in and out of that bulge. That bulge is always there. However. And it's always in the same place. Well, it's a little bit ahead of the moon in the moon's orbit, right? But we rotate faster than moon's orbit around us. So we rotate into and out of the tidal bulge. And because the moon still moves around us, high tide shifts each day by about 45 or 50 minutes because of this effect. It's all one synchrony of time and space. Because of the effect of the tidal bulge being slightly ahead of us. It's slightly ahead of the moon.

But the moon is orbiting the earth. So how far does the moon go in a day? It goes about 45, 50 minutes worth of your clock time around the earth. So therefore, the tidal bulge has shifted by that much. But that's not even my point. The bulge is always there. When we line up with the sun, the sun's high tide adds to the moon's high tide. and you get full moon high tides. So when people think the full moon has some extra tidal forces on us, no. It's the sun lining up. It's the sun added to the moon's tide, and the moon's tide is just always there. But the sun adds to it. And so it's a delusion. We think, oh, the rays of the sun, the moon. The sun. We have werewolves. We have all life-canthropic stories that have been written. We've got all of this about moonlight.

And plus the moon is only reflected sunlight. So if the werewolf turns into a werewolf under full moon, it should definitely do it under sunlight. And a lot faster. And with a lot more hair, bigger teeth, bigger fangs. So what does remind us the full moon high tide, which means sort of the sun and the moon have sort of aligned, that happens every how often? What is the frequency of that? Once a month. Well, actually twice a month. So it'll work for Full Moon because Earth, Moon, and Sun are lined up and it works for New Moon where the Moon is on the other side and you get the tidal bulge. So you get the highest of the high tides during New Moon and Full Moon. So people aren't thinking that the New Moon has special powers even though it's making tides just as high as Full Moon because you can't see the New Moon. So people are very influenced by what they see and what they think is true rather than all that they can't see.

And what we otherwise know is true. True. Yeah, there's a lot there. Hello from Flesherton, Ontario. Not going to be a part of the 51st State Canada. But we need your health care. We feed your health care, your pleasant qualities, your lovely manner with people, your maple syrup. We've only got a couple of states that really generate that. Come on. Well, Vermont, New York. That's in some New Hampshire, yeah. There's been some Maine too. Maine too, yeah. Neil, my 13-year-old son was so blown away by our experience at your show, he wanted me to pass on this question. Why are your t-shirts so expensive after the show? No, that's not the question. I have no merch. People said I should, but there's no merch. You sell those Neil DeGrasse Tyson fake mustaches for 15 each. I know you do. Neil, my 13-year-old son was so blown away with our experience at your show, he wanted me to pass on this question. Could you ask if he thinks your memories and consciousness are made from atoms. Love you guys and keep the laughs coming, Andrew Nisker. So there's my favorite joke that's safe for elementary school. Never trust atoms. They make up everything. That's safe for like third graders. We're good there. So yeah, I mean, We so want to believe that there are forces and things going on that defy account or description or investigation by the laws of physics. That urge is very strong because you'll never, you might explain language or whatever, but you'll never explain love. And I say, is it because you don't want us to ever explain love? Would you have some reason for thinking that this is really true?

a couple of things about our consciousness or our thoughts. Let's just talk with our thoughts. We have come to learn that the brain operates via electrochemical signals between the synapses of our nerve cells, of the brain cells in your head. Okay. And there it is. By the way, the medical profession could not have known that before the discovery and the harnessing of electricity. Think about that. So much of physics has fed the medical profession's powers to diagnose and treat the human body, that if anyone says, okay, we're done with physics, now let's invest all in medicine, there's a limit to how far that'll go given how much they've tapped discoveries by my people in my fields.

Okay. So when you know that thoughts are communicated through electrochemical signals and then you stick a probe in there and stimulate a

part of the brain that you know behaves in a certain way, and then you see the person react, you kind of know what's going on there. This is as it is. Yeah, atoms are the building blocks of cells and the cells of the building blocks of our neurophysiology. Okay. So, but the real evidence here, because you can't just experiment on people's brains that, you know, we can't do that. Hang on a second. All right. Let him go. We can't let your kids, your kids in the box. Yeah. Get them out of the box and give them some peanuts and hit them. We can't do the experiment. Go ahead. Sorry. People who have many strokes, right? So a stroke of broken blood, blood vessel, this oxygen doesn't get to part of the brain, if it's mini-strokes, the identity of the person begins to fade, depending on what part of the brain was touched by the stroke. And so is it their facial recognition? Is it their spatial recognition? Is it their appetite for food? Is it their ability to speak? Is it their hearing, sight, locomotion? If you have a series of mini-strokes bits and pieces of you, basically disappear. So everyone who says what happens to your consciousness when you die, a series of many strokes or many deaths of your brain. And as far as we can tell, that's what's happening to you when you die. None of that works. So now that's not very comforting to be especially not to religious people have a whole story of what happens to you when you die.

All I'm saying is, if you're going to ask a scientist to answer that question, the point is, once your brain ceases to undergo electrochemical changes, then you don't exist anymore. Everything we know about you no longer exists. So, I mean, dementia is sort of in that family, right? For example, correct. And that's not even a stroke. I should have included that. But it's in that family of, you know, so my mother has it right now and you can see there's just pieces of it. but the fascinating thing is her long-term memory is insanely accurate. It's stored differently and possibly even in a different place. I would introduce you to her and 10 minutes later she'd be like, how do you know my son again? But then she can in the same breath relay a story when she was 12 of spending a day with her father with great exactitude. And so Is it the shifting of the atoms in the cells? Is it the shifting of the atoms in the cells that's happening there? Is it the death of some atoms in cells? The atoms don't have life or death. They just comprise larger molecules. And the cells are composed of the molecules. So you can't

blame atoms directly. You have to credit or blame the functioning of living tissue.

Okay, that's really what it is. The atom is just an atom. Now, if I may, I mean no disrespect, but I heard this joke told by another comedian who said that if you're a comedian, being around people with dementia is excellent because you could tell the same joke every 10 minutes. Absolutely. People with short-term memory dementia. It also helps with guilt because like you can say to your mother, well, I haven't seen you in a while. No, I was just here yesterday. Oh, that's right. You know what I mean? So like, you don't have to feel so bad. Yeah, exactly. It's like, oh, you're such a good son. I know. I was here yesterday and I painted the whole house. The point is Adams, everything that is not empty space or...

Energy in the form of photons represents as particles in the universe and certain combinations of particles we call atoms if you have a nucleus and at least one electron it's an atom and so Adams comprise everything That's that's hence the joke don't never trust Adams. They make up everything If you go down to like a snowflake, and a snowflake is six-sided snowflake, that's a crystalline structure of the water molecule. Water is H₂O. You break that apart. You have two hydrogen atoms, one oxygen atom. So atoms are everywhere. But atoms don't have agency. They just, well, sorry, they might want to combine with other atoms to make molecules. But that's, it's a very simple, are we mating or not?

It's not otherwise, let me give Paul Mercurio dreams. That happens based on life experience and leftover neurosynaptic history of your brain and stuff that's unresolved, trauma that might have happened in your life, joys. So yeah, atoms are everywhere. So Paul, we got enough time for like a lightning round. This is Todd Chambers from Yuba City, California in the system of the universe. What is or are the purposes of a black hole? Do they recycle matter and energy? To speak of purpose implies intent. And in physics, we generally don't think of the universe as having intent. They're just laws and forces and things flow according to those forces, guided by the laws of physics as we've come to describe them. So you can ask what role they play.

And if you wander too close to a black hole, that's all she wrote. You'll get pulled in, spaghettified, and you won't be able to tell anybody about it. So to think of a role, no, there are no roles. There's just what it does. And by the way, if the sun became a black hole today, Earth would plunge into darkness, but we can still continue to orbit around it. in just the same way we are right now. We wouldn't be pulled in? Yeah, black holes are not giant sucking machines. But there's stronger gravitational force. Only at its surface. Don't disagree with me. You don't know what you're talking about. I'm not disagreeing with you. I'm just simply saying you're wrong. There's a difference. Two politicians will disagree. Yes. Okay.

But if one person is wrong, you're just wrong. It's not that we disagree. You're just flat out wrong. Yeah, yeah. So yeah, so it doesn't become a giant sucking machine. We speak of the high gravity of a black hole because of the high surface gravity. Because the black hole is so small, you get really close to its center. And that's where the extreme gravitational phenomenon occurs. Got it. Okay. All right, we'll keep going here. Liam Corcoran. Hi, Neo. This is Liam from Rhode Island, my home state, the best little state in the Union. Is there anywhere in the universe that you could be outside the gravitational influence of any matter besides your own? And would this be the closest you could get to experiencing pure space time, totally unwarped by matter? I love that question. I love that question.

So there is, oh, I forgot what the metric is called. There are approximations we make when we're doing some calculations. We assume space has no matter or energy in it, so that it's perfectly flat. And then you can do some calculations more easily than if you admit that there are stars or planets or black holes distorting that spacetime. The mathematics becomes much more complicated if the spacetime is curved than if it's flat. So now, in 1968, if memory serves, Walter Cronkite, the anchor for CBS News, was describing in December the voyage of Apollo 8 to the moon. And he says, as of 9.22 this afternoon, Apollo 8 astronauts have left the gravity of Earth. And I remember hearing this, like, what? They're on their way to the moon. And last I checked, the moon orbits Earth. So I know what he meant.

But that's not what he should have said. What he meant was we reached a point where the moon's gravity, pulling it in, is stronger than our gravity

trying to come back to Earth, so that the moon's gravity takes over and it just falls towards the moon. Earth's gravity extends to infinity as does every other gravitational field. You cannot escape the extent of gravity, no matter what. But it weakens over time. The farther you get away from Earth, it dissipates in strength. And weaker it is, but it's never zero. That's true for everything, the sun, the galaxy, galaxy clusters, everything. My name is Emily from Berkeley. I am a first-time poster. I hope this is an interesting question. Many times when people talk about particle pairs being separated at outer surface of black holes, where one half of the pair is on the outside and one half is doomed to the inside. They will say that it should be theoretically possible to reconstruct the information that fell into the black hole. Why is it important to theoretically retain this possibility? Oh, this comes from information theory because there's a certain amount of information. And if you fall into a black hole, was the information destroyed?

That's the question and so there's a big debate about information information theory is relatively new, you know, it's sort of Well, it comes out of entropy. So in the last hundred years or so Especially in the last 50 50 years puts it in the 1970s. I guess. Wow. Yeah And if you believe or want it to be so that information is never destroyed, then when the black hole later evaporates, which it will do by Hawking radiation, well, that is, excuse me, that is the way it evaporates. Okay. So, so that particle pair is conjured out of the gravitational field. And the, if you took an inventory of the particles that appear out of the gravitational field of the black hole, they will exactly match everything the black hole has ever eaten in its life. How so? Well, what's freaky is it's not the same particle coming out that survived the trip in and survived the trip out, particle being created from the energy field of the black hole.

Energy is creating it and it has a memory of what was there before that's tandem out to preserving the information that went in so Now there's a little bit that I'm unsure of we have to bring Jana Levin on this If you throw in a DNA molecule, there's a lot of information. There's your whole identity It's not going to come back out as a DNA molecule in particle pairs. No It'll come back altered in some way now. It won't come back as a molecule at all. It'll come back as particles. So how do I recover the information that the

DNA molecule contained as a representation of you? I don't have an answer for that. We've got to check with either one of the two Bryans or with Gen 11 on that. We'll bring them back on. Don't worry about it. I got it. I got it. You got this. Yeah.

What happens is it goes in and it's centrifugal force that happens along with a certain level of radiation red wave and then you get shit thing. All right, moving on. Okay. Mercedes Dominguez. Hello, Mercedes here from Denver, Colorado. I've heard of the Cosmic Web before. My question is, is that a real thing? And if so, what is it? Yeah, so it's just a word that we apply to how we find the galaxies configured in the universe. So galaxies are not evenly separated. There's gravity operating, and there was some fluctuations in the early universe that preloaded some parts of the universe relative to others. When you step back and take a look at it all, you see this sort of network of filaments and clusters and It says, though, you took a sponge, cut it in half, like a nice thick sponge, cut it in half, and look at the textures and structures. They're places where the fibers meet. They're other places where it's empty. It's almost like a honeycomb in a way, not as symmetrical. Yeah, but not as organized, not as structured. Correct. So the galaxies aren't sprinkled evenly. They line up along huge strands of matter, mostly dark matter, with these empty voids in between. And it looks like a...

three-dimensional spider web in a way. Or a sponge, yeah. I mean, cosmic web sounds a little more romanticized than cosmic sponge. Yeah, exactly. Yeah, that's a little, that's a tough sell to the, yeah. But the point is the universe isn't random, it's organized. I mean, it's not like Pinterest organized. I wouldn't even use the word organized. I mean, gravity collects it. You want to call that organization. It's like a garage with stuff piled into the corners of every possible spot. That's a good point. In fact, here's an interesting phenomenon. If you look at a movie where they show a scene in the fall, basically they're set designers scattering leaves onto the street, if it's not actually in the fall. What they always get wrong is they try to scatter the leaves evenly on the street. It's never that.

The breeze comes by and it plows into one corner. Some other new leaves fall down, but it's never evenly distributed. You can always tell because people don't understand random. They think random is scattered

everywhere. Random has clumps. That's how that is. So I can always tell if they fake the street for that reason. But science can't answer the... The randomness question here, right? Isn't that to me the fascinating thing here? It's sort of in a way organized chaos. Maybe that's not the right technical term. No, no, no. Forces of gravity are operating on it. So there's nothing more mysterious than that. And dark matter is mysterious. We don't know what that is, but that's operating on the gravitational fields around which all of this is collecting. So we speak of that entire map.

of the universe as a cosmic web. So Paul, that's all the time we have. Plus, I think I hear your mother calling you for lunch. It's great to be on with you. Yes, man. And keep us posted on the end of Colbert or the end of days. Oh, you'll know because I'm going to be coming out to your neck of the woods. I'm doing landscaping after this. OK. I need the money, buddy. Can you buy a place and get a lawn so I can put it? Come on, be a friend. Be a friend, will you, Leo? Get a lawn. All right, we'll work on it. All right. All right, Paul McHurio, always good to have you, man. Yeah, always great. Chubby Terry Crews. This has been another installment of Cosmic Queries Grab Bag Edition with...

Paul Mercurio, Neil deGrasse Tyson, as always bidding you to keep looking up.